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Literature Review of Units of Caring, Pain & Suffering Measurement, and Aggregation

Quantifying Caring and Suffering: Early Concepts

Early utilitarian philosophers attempted to formalize units of pleasure and pain. **Jeremy Bentham (1789)** introduced the *felicific (hedonic) calculus*, proposing that a decision's morality could be determined by summing up the **"units of pleasure" and "units of pain"** it produces. In Bentham's framework, a *hedon* was considered a unit of happiness (or pleasure) and a corresponding unit of pain was sometimes called a *dolor*, allowing a theoretical calculation of net utility for any action. This notion that happiness and suffering could be quantified was foundational for utilitarian ethics, though it remained more conceptual than practical.

In colloquial and modern contexts, people have sought more tangible "units" of caring. **Eliezer Yudkowsky (2007)** argued that **"money is the unit of caring,"** suggesting that how much someone is willing to spend is a concrete measure of how much they care about an outcome. In other words, **preferences reveal themselves in monetary trade-offs** – if one wouldn't spend \$5 on a cause but would on a latte, it implies the latte is valued more than that cause. This tongue-in-cheek argument underscores the idea that scarce resources (like money) can serve as a rough proxy for caring or utility in the real world. Similarly, the effective altruism movement quantifies altruistic impact in **lives saved or quality-adjusted life years per dollar**. For example, providing insecticide-treated bednets in poor regions can **save a life for about \$5,000**, far cheaper than in rich countries. Such metrics illustrate the concept of a "unit of caring" in practice – one donated dollar can be mapped to a fraction of a life saved or suffering averted, enabling comparison of interventions.

Measuring Pain and Suffering

Pain is inherently subjective, which makes it challenging to measure and quantify. The International Association for the Study of Pain (IASP) famously defines pain as *"an unpleasant sensory and emotional experience"* – emphasizing that it is **a subjective experience, not just a physical signal**. As a result, **self-report is the gold standard** for assessing pain intensity. Clinically, simple **numeric rating scales (0–10)** are commonly used, where 0 = no pain and 10 = worst pain imaginable, to capture a patient's pain in a single number. While easy to administer, these scales compress a complex experience into one dimension and assume everyone's "10/10" is comparable.

Researchers have long sought more nuanced or objective measures. In the 1940s, **Cornell University scientists** attempted to create an **objective unit of pain called the "dol."** They developed a dolorimeter device to apply controlled noxious stimuli and defined a scale from 0 to 10 "dols" of intensity. Notoriously, they even tested pain on medical students and women in labor by **burning the skin** between contractions to calibrate this scale. One **dol** was intended to represent a just-noticeable difference in pain, and they claimed (controversially) that pain intensity summation was arithmetic – e.g. four 2-dol stimuli = an 8-dol pain. However, **the "dol" unit never caught on**. Other researchers could not reproduce the exact measurements, and the approach was abandoned as **pain's subjective qualities resisted such linear objectification**. This episode highlighted that pain isn't like weight or temperature – it cannot be fully captured by a single objective unit.

Instead of pure intensity scales, multidimensional self-report tools were developed. **Ronald Melzack's McGill Pain Questionnaire (1975)** was a breakthrough in measuring pain's complexity. It asks patients to pick words describing their pain's quality and affect (e.g. *"sharp," "throbbing," "sickening," "tiring"*), grouping them into **sensory, affective, and evaluative categories**. The MPQ produces both a pain rating index and qualitative profile, acknowledging that **pain has multiple dimensions** – not just intensity, but also the kind of sensation and the distress it causes. This was built on earlier ideas that pain involves both a discriminative sensory component and an emotional-motivational component. Similarly, **Melzack & Wall's Gate Control Theory (1965)** showed that pain signals are modulated in the spinal cord and brain by psychological factors, underscoring that **pain perception is not purely stimulus-driven** but shaped by context, attention, and mood. Later, Melzack proposed a brain "neuromatrix" for pain integration, explaining phenomena like phantom limb pain (pain in absence of injury). All these insights reinforced that measuring pain requires accounting for subjective experience and interpretation by the brain.

Despite these complexities, the medical field often needs to quantify pain and suffering for practical reasons (tracking treatment outcomes, disability claims, etc.). **Behavioral and physiological proxies** – such as facial expressions, heart rate, or neuroimaging – have been explored to estimate pain when self-report is unavailable (e.g. in infants or unconscious patients). For instance, recent advances in neuroimaging have identified patterns of brain activity that correlate with pain, sometimes dubbed "*pain signatures*." Researchers like Wager et al. (2013) even developed an fMRI-based model that could predict pain vs. no-pain conditions with high accuracy. This raises the question: could brain scans serve as an **objective "pain-o-meter"**? In 2015, a U.S. court **allowed fMRI evidence of pain** in a chronic pain legal case, reflecting the growing interest in neuroscience-based assessment. However, a **2017 international task force (IASP)** urged caution: they concluded that **no brain-based test can yet replace self-report**, and ethical and legal standards for admitting such evidence must be stringent. They emphasized that pain's subjective nature means any objective test will have uncertainty; using brain scans to judge someone's "truthfulness" about pain could be problematic. In sum, while technology can augment pain measurement (for example, by identifying when someone with communication barriers is likely in pain), **the first-person experience remains the cornerstone**. Pain assessment today often combines approaches – self-report scales for subjective intensity, questionnaires for qualitative aspects, and careful observation of behavior or physiology – to form a fuller picture. But any purely empirical model must grapple with the fundamental privacy of suffering: only the sufferer truly knows its extent.

Modeling Pain and Suffering: Neuroscience and First-Person Perspectives

Understanding suffering requires bridging the gap between **third-person neuroscience and first-person experience**. Modern neuroscience has mapped many neural correlates of pain and emotion – e.g. activity in the **anterior cingulate cortex and insula** reliably increases during painful experiences, and even during empathy for others' pain. Yet, **correlation is not identity**: brain activation alone doesn't capture the *felt quality* of pain. Researchers are increasingly aware that a **purely neurocentric model is incomplete**. Thacker and Moseley (2012) argue that **"modern neuroscience [does not] thoroughly consider pain phenomena and experience"**, suggesting that a purely objective account misses crucial subjective elements. They call for a *first-person neuroscience* – integrating insights from phenomenology (the careful description of lived experience) into scientific study. In essence, science may need philosophy or introspective methods to precisely understand pain and suffering.

One approach to integrating these perspectives is by proposing theoretical frameworks that include a model of the self. **Philip Gerrans (2024)** offers an *active inference model* in which **pain is deeply linked**

to the brain's model of the self. He suggests that **pain experience emerges from distributed processing anchored by a "multidimensional self-model."** In conditions like pain asymbolia (where a patient feels pain but isn't bothered by it), Gerrans explains the dissociation by different levels of self-modelling: the sensory pain signal may be intact, but the higher-level attribution "this is *my* suffering" is disrupted. His account treats pain as a type of **allostatic control signal** – part of the brain's predictive regulation of the body's needs. Suffering, in this view, arises when pain is interpreted in the context of the person's goals, memories, and identity (the "self-model"). Such models tie together neuroscience (hierarchical brain processes) with first-person structure (the sense of self) to explain why pain feels the way it does and how it can vary. They also align with the idea that **suffering is not just raw sensation but "pain interpreted" by the mind**.

Another line of work examines **empathic pain versus direct pain** to probe the components of suffering. **Zaki, Singer, Keysers et al. (2016)** proposed that when we see someone else in pain, the brain generates an *empathic response* that shares some, but not all, elements of actual pain. Using neuroimaging and careful experiments, they found that **empathizing with pain activates core pain regions (insula, ACC)** that represent the *unpleasantness* (the negative affect) in both self and other's pain. However, certain specifics – like the precise sensory location/intensity – are only present during first-hand pain. This led to a **layered model of pain**: some brain circuits encode a **general suffering component ("this is bad")** common to various aversive experiences, while other circuits encode the **specific pain component ("my hand is burning")**. Such findings reinforce a theme: suffering arises from an interaction of multiple levels – sensory-discriminative signals, emotional evaluation, cognitive context, and self-relevance.

The **mind's interpretation and context** can amplify or diminish suffering even if pain signals remain the same. For instance, a soldier wounded in battle may report little pain (context and meaning modulate suffering), whereas catastrophic thinking in chronic pain can worsen the felt suffering without increasing noxious input. First-person accounts and cognitive models highlight factors like **attention, beliefs, and control** in shaping suffering. Psychological research on *pain catastrophizing* shows that how one *thinks about* pain (e.g. feeling helpless or believing it will never end) strongly affects pain intensity and long-term outcomes. This implies any comprehensive model of suffering must incorporate **subjective appraisals and high-level cognitive states**.

In summary, current models of pain and suffering are converging on a view that combines **neuroscience with first-person perspective**. Suffering is seen as neither a purely physiological signal nor a purely abstract feeling, but a complex phenomenon arising from brain-body systems *and* personal context. Ethical implications follow: if we understand the mechanisms of suffering, we might better alleviate it (through therapies addressing both neural and psychological aspects). At the same time, acknowledging the first-person perspective reminds us that **subjective reports are indispensable** – no brain scan or algorithm can tell us someone's inner experience with complete certainty. As neuroethicist Karen Davis notes, **respecting the privacy of subjective experience is crucial even as science advances**, to avoid overstepping into "mind-reading" without individuals' consent.

Aggregating Suffering and Well-Being

Beyond individual experience, researchers and policymakers face the challenge of **aggregating pain, suffering or well-being across groups**. In healthcare and global health, this has led to standardized metrics that attempt to sum up suffering averted or well-being added by an intervention:

- **Quality-Adjusted Life Year (QALY):** The QALY is a composite index that combines **length of life with quality of life** in a single number [1]. By convention, **1 QALY equals one year of life in perfect health** [1]. A year bedridden in moderate pain might count as, say, 0.5 QALY. If a treatment yields an extra year of healthy life for a patient, that's 1 QALY gained; if it restores a patient from 50% health to full health for one year, that's 0.5 QALY gained. QALYs allow different health outcomes to be compared on a common scale of "health-equivalent years." They are widely used in cost-effectiveness analysis: e.g., a new drug might cost \$50,000 per QALY gained. This helps health systems allocate resources to maximize the total QALYs (and thus minimize total suffering/ill-health). The QALY framework essentially treats **health as accruing "utility" over time**, where being alive and healthy yields full utility and death yields zero [1]. (States worse than death can even be assigned negative QALYs in some models.)

- **Disability-Adjusted Life Year (DALY):** Conceptually the inverse of a QALY, a DALY measures **health lost** rather than health gained. **1 DALY corresponds to one lost year of healthy life** due to disease, disability, or premature death. The World Health Organization defines total DALYs for a disease as the sum of: years of life lost (if someone dies young from it) + years lived with disability (time spent in a state of reduced health, weighted by severity). **DALYs thus quantify the burden of diseases** on populations. For example, if a population has 100 people each living 10 years with a condition rated at 50% health, that's 500 DALYs (50 years of healthy life lost in total). Global Burden of Disease studies use DALYs to compare the total suffering caused by different conditions – e.g., in 2019, low back pain caused tens of millions of DALYs globally, making it one of the top causes of disability. By aggregating in this way, public health officials can prioritize interventions that save the most DALYs. One DALY can be thought of as one unit of suffering (one full healthy year lost) to be averted.

Both QALYs and DALYs face **ethical and methodological debates**. They require assigning numerical "quality weights" to health states (e.g., is a year with chronic pain worth 0.7 or 0.4 of a healthy year?). These weights often come from surveys of patients or the general public, and results can vary. Critics also note that such aggregation inherently involves **value judgments** – e.g. is extending many lives by one year better or worse than saving one young person's life outright? However, despite imperfections, QALYs and DALYs are **widely-used empirical measures** that allow pragmatic comparisons across diseases and programs. They represent perhaps the closest realization of a "unit of caring" in policy: a life-year of health gained or lost.

In effective altruism and welfare economics, scholars also discuss aggregating well-being more broadly. Some propose using **subjective well-being scores** (like average life satisfaction on a 0–10 scale) as societal metrics, effectively summing self-reported happiness across people. Others explore monetary proxies: e.g. **"Value of a statistical life" (VSL)** is an economic construct where peoples' willingness to pay for small risk reductions implies a dollar value for saving one life (often in the range of \$1–10 million in rich countries). While the VSL is used in regulatory cost-benefit analysis, effective altruists focus on the much lower cost per life saved via global health interventions as noted earlier. The differences highlight **global disparities in the "cost of suffering"**: spending \$5,000 on malaria prevention in Africa might save a life (equivalent to many QALYs gained), whereas the same amount in a developed country would achieve only a fraction of a QALY by, say, funding an expensive treatment.

Another important aspect is **aggregation over time and individuals** in ethical theory. Avoiding deep theoretical tangles of population ethics (which the present work sidesteps), empirical aggregation typically takes individuals as units and adds up their experienced outcomes. This can be seen in projects like the **World Happiness Report**, which reports country-level happiness by averaging thousands of people's self-ratings. It implicitly assumes one person's point on the scale is comparable to another's. Likewise, pain intensity studies sometimes try to compare pain across patients – e.g., is one patient's "8/10" similar to another's? In clinical trials, researchers aggregate pain outcomes by looking at group means or the proportion achieving a certain reduction, effectively summing or averaging individual pain reductions to judge a treatment's efficacy.

Crucially, **aggregation treats units of suffering or happiness as interchangeable and additive**. This is useful for big-picture decisions, but it also draws criticism. Some argue that suffering is personal and can't be truly "added up" across people – the utilitarian sum might ignore distribution (e.g., is it better for 100 people to each suffer a little, or 1 person to suffer a lot? The sums could be equal but ethical intuitions differ). Nevertheless, in practice, public health and humanitarian work operate on aggregated metrics: e.g., "this program will avert 10,000 DALYs next year" or "this policy could save \$3 billion in lost productivity." These statements guide policy by translating suffering into numbers.

In **summary**, the literature shows a progression from philosophical musings on units of pleasure/pain, to attempts at scientific measurement of pain, to modern quantitative frameworks for health and welfare. **"Units of caring"** can refer to money, health units like QALYs/DALYs, or even theoretical hedons – but all share the goal of making caring and suffering **measurable and comparable**. Each approach has limitations: **subjective experiences resist full quantification**, and any unitization inevitably simplifies a complex reality. The challenge for current researchers is to improve these measures (through better neuroscience, psychology, and economics) while remembering the qualitative depths behind the numbers. By combining first-person perspectives, scientific measurements, and ethical reasoning, we inch closer to a comprehensive understanding of suffering – one that may eventually inform better alleviation of pain and more effective caring in the world.

References (Vancouver Style with Summaries)

1. **Bentham J. (1789). *An Introduction to the Principles of Morals and Legislation*.** – Bentham proposes that pleasure and pain can be quantified and summed (a "hedonic calculus"), treating morality as an arithmetic of units of happiness vs. suffering.

2. **Yudkowsky E. (2007). "Money: The Unit of Caring." LessWrong.** – An essay arguing that how we spend money reveals what we care about; *"to a first approximation, money is the unit of caring"*, meaning willingness to pay is a proxy for the value one places on goals or causes.

3. **Veronese K. (2012). "Scientists created a pain measurement scale by burning the hands of women in labor." *Gizmodo*.** – Describes 1940s research that defined the **"dol"** as a unit of pain using a 0–10 scale of applied heat; although innovative (and unethical by today's standards), the **dol system was not reproducible or adopted**, highlighting the difficulties of objectifying pain.

4. **Hardy JD, Wolff HG, Goodell H. (1940s). *Studies on Pain* (multiple papers).** – Pioneering experiments using a **dolorimeter** to measure pain thresholds and intensities; introduced the term **"dol"** for a unit of pain and a 21-point pain scale, but later research failed to validate this linear scale, leading to its abandonment. *(Original experimental evidence of attempts to quantify pain objectively.)*

5. **Melzack R. (1975). “The McGill Pain Questionnaire: Major properties and scoring methods.” *Pain*, 1(3):277-299.** – Introduces the **McGill Pain Questionnaire (MPQ)**, which uses 78 descriptive words to capture sensory, affective, and evaluative dimensions of pain; a key tool demonstrating that **pain has multiple qualitative dimensions** beyond intensity, allowing systematic measurement of pain experience in clinical research.

6. **Wall PD. (1999). *Pain: The Science of Suffering*.** – A comprehensive overview of pain mechanisms by a co-developer of Gate Control Theory; explains how pain signals are modulated by the nervous system and the mind, and argues that **suffering is constructed by the brain**, not simply transmitted from injured tissue. *("Pain is not just a message but a complex experience shaped by attention, emotion, and past experience.")*

7. **Thacker MA, Moseley GL. (2012). “First-person neuroscience and the understanding of pain.” *Med J Aust*, 196(6):410-411.** – A commentary asserting that current neuroscience approaches do not fully capture the **subjective, first-person aspect of pain**; calls for integrating philosophical and phenomenological insights with neuroscience to achieve a more **precise understanding of pain experience**.

8. **Gerrans P. (2024). “Pain, suffering and the self: An active allostatic inference explanation.” *Neuroscience of Consciousness*, 2024(1):niae002.** – Proposes that pain is processed within a **hierarchical self-model** in the brain; explains conditions like pain asymbolia by showing how normal pain’s emotional impact requires linking the signal to the self, and frames suffering as the result of the brain’s predictive regulation (allostasis) when the self perceives threat or harm. *(Offers a theoretical bridge between neural mechanisms and the lived experience of suffering.)*

9. **Zaki J, Wager TD, Singer T, Keysers C, Gazzola V. (2016). “The Anatomy of Suffering: Understanding the relationship between nociceptive and empathic pain.” *Trends Cogn Sci*, 20(4):249-259.** – Reviews research on **empathic pain**, proposing that observing others’ pain activates general “suffering” circuits (affective pain components) in the observer’s brain **shared with firsthand pain**, but with differences in sensory-specific processing; presents a model where pain experience consists of both shared components (unpleasantness, alarm) and unique components (localized sensation, self-versus-other).

10. **Davis KD *et al*. (2017). “Brain imaging tests for chronic pain: medical, legal and ethical issues and recommendations.” *Nat Rev Neurol*, 13(10):624-638.** – A consensus statement by pain and neuroethics experts evaluating fMRI and other brain-based “pain detection” methods; concludes that **self-report remains the gold standard due to pain’s subjectivity**, and while brain imaging can provide supportive evidence, it **should not yet be used as a sole lie-detector or objective arbiter of pain** in legal/clinical settings. *(Emphasizes caution in using neuroscience for pain assessment, highlighting ethical implications.)*

11. **WHO. (2021). *Global Health Estimates: Disease burden*.** – Defines the **Disability-Adjusted Life Year (DALY)** as **“one lost year of ‘healthy’ life”** and explains that total DALYs are calculated by summing years of life lost and years lived with disability for a population; used to compare and aggregate the burden of diseases and injuries in epidemiology, effectively quantifying total suffering (health loss) across different groups.

12. **ICER (Institute for Clinical and Economic Review). (2020). “Cost-Effectiveness, the QALY, and the evLY.”** – An overview of how **Quality-Adjusted Life Years (QALYs)** are used in health economics; affirms that **one QALY equals one year in perfect health** and that QALYs have been

the standard outcome metric in cost-effectiveness analyses for decades. *(Supports the use of QALYs as a measure to aggregate improvements in longevity and quality of life from medical interventions.)*

13. **Effective Altruism Handbook. (2022). “Providing basic medical supplies in poor countries.”** – Illustrates the concept of **cost per life saved** and **cost per QALY** in global health: reports that according to GiveWell’s research, **distributing insecticide-treated bednets can save a child’s life for about \$5,500** (roughly equivalent to buying ~30 QALYs, given one life saved corresponds to decades of healthy life). This highlights how quantifying impact in terms of lives or QALYs allows comparison of charitable interventions and embodies a practical “unit of caring” for altruistic decision-making.

14. **World Happiness Report. (2023). *Chapter 2: Well-being over time*.** – Uses aggregated **life satisfaction scores (0–10)** across countries to measure social well-being; explains that national average happiness is computed by surveying thousands of individuals, effectively treating one life-satisfaction point for one person as equal to the same for another. *For example, Finland’s average life satisfaction of ~7.8/10 (for 2021–2023) is the highest globally, indicating a high aggregate of positive subjective well-being.* (Demonstrates an alternative way to aggregate “positive experience” or lack of suffering at the population level, complementary to health metrics.)

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[1] Quality-adjusted life year - Wikipedia
https://en.wikipedia.org/wiki/Quality-adjusted_life_year